MEASUREMENT
NOT SENSITIVE

MIL-PRF-3122J
24 October 2000
SUPERSEDING
MIL-PRF-3122H
17 January 1996

PERFORMANCE SPECIFICATION

LEATHER, CATTLEHIDE, FOR FOOTWEAR

UPPERS AND GUSSET, CHROME TANNED, FATLIQUORED

This specification is suggested for use by all Departments and agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This specification covers smooth finish, full grain and corrected grain, chrome tanned, fatliquored, cattlehide leather for footwear uppers and gusset.

1.2 Classification. The leather will be of two types in the following treatments as specified (see paragraph 6.2):

Type I – Leather, Cattlehide, for Footwear Uppers, Chrome Tanned, Fatliquored
Type II – Leather, Cattlehide, Gusset, Chrome Tanned, Fatliquored (see paragraph 6.3)

Treatment A – Treated for resistance to water absorption
Treatment B – Treated for resistance to water penetration
Treatment C – Not treated for resistance to water
Treatment A/B - Treated for resistance to water absorption and penetration

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Supply Center Philadelphia, Clothing and Textiles Directorate, Attn: DSCP-CRFD, 700 Robbins Avenue, Bldg. 6D, Philadelphia, PA 19111-5096, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8330

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2. APPLICABLE DOCUMENTS.

2.1 General. The documents listed in this section are specified in Sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in Sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents will be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see paragraph 6.2).

2.3 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted will be those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see paragraph 6.2).

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

Chromatic Transference Scale

(Copies may be obtained from the American Association Of Textile Chemists and Colorists, P.O. Box 12215, Research Triangle Park, NC 22709.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D1610 – Standard Practice for Conditioning Leather and Leather Products for Testing

D2098 – Test Method for Dynamic Water Resistance of Shoe Upper Leather by the Dow Corning Leather Tester

D2099 – Test Method for Dynamic Water Resistance of Shoe Upper Leather by the Maeser Water Penetration Tester

D2212 – Test Method for Slit Tear Resistance of Leather

D2617 – Test Method for Total Ash in Leather

D2807 – Test Method for Chromic Oxide in Leather (Perchloric Acid Oxidation)

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D2810 – Test Method for pH of Leather Chemical Tests

D2868 – Test Method for Nitrogen Content (Kjedahl) and Hide Substance Content of Leather

D3495 – Test Method for Hexane Extraction of Leather

D3790 – Test Method for Volatile Matter (Moisture) of Leather by Oven Drying

D4705 – Test Method for Stitch Tear Strength of Leather, Double Hole

D6012 – Test Method for Determination of Resistance of Leather to Color Stain Transfer

D6014 – Test for determination of Dynamic Water Absorption (Leather surfaces)

D6015 – Test Method for Static Water Absorption of Leather Surfaces

(Copies may be obtained from the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428.)

(Non-Government standards and other publications are normally available from the organizations that prepare and distribute the documents. These documents also may be available in or through libraries or other informational services.

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS.

3.1 Material. The leather shall be full grain or corrected grain, chrome tanned cattlehide sides.

3.2 Staining. Except as noted below, the flesh side of the leather shall be tested for staining (Bleeding) in accordance with paragraph 4.3. The stain on the test pad shall have an expanded AATCC value of not less than 3.

Note: Leather which has a natural colored flesh side that has been subject to small amounts of dye added to the drum in order to level the shade of the natural color need not be tested for staining.

3.3 Finish. Corrected grain leather shall be buffed or snuffed only to the extent that is necessary to remove superficial blemishes. The grain side of the leather shall have a light application of finish containing only sufficient pigment to assist in making the color uniform. The finish of the grain side shall be smooth.

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Note: A fine hair cell embossed pattern may be applied to the grain surface of the side. Printed leather shall not be permitted. If embossed, the leather shall match the standard sample for fine hair cell pattern. Leather previously dyed or finished in a color not specified and then redyed or refinished a specified color shall not be acceptable.

3.4 Fungicide. Unless otherwise specified (paragraph 6.2), one of the following leather fungicides is required.

3.4.1 TCMTB. The leather shall contain not less than 100ppm and not more than 400ppm of active ingredient 2- (thiocyanomethylthio) benzothiazole (TCMTB), (see paragraph 4.4.1).

3.4.2 Ortho- Phenyl Phenol. The leather shall not contain less than 540 ppm and not more than 2580 ppm. (see paragraph 4.4.2)

3.4.3 Diiodomethyl para Tolylsulfone. The leather shall contain not less than 300 ppm and not more than 2000 ppm. (see paragraph 4.4.3)

3.5 Water resistant treatment. When treatment A, B or A/B is specified, the leather shall meet the following requirements as applicable.

3.5.1 Treatment A (water absorption). When treatment A is specified, specimens of the leather shall gain not more than 30 percent of their weight by water absorption when tested as specified in paragraph 4.3.

3.5.2 Treatment B (water penetration). When treatment B is specified, at least 80 percent of the specimens tested shall withstand not less than 15,000 cycles by ASTM D2099 or not less than 4,000 cycles by ASTM D2098 without water penetration, when tested as specified in paragraph 4.3.

3.5.3 Treatment A/B (water absorption/penetration). When treatment A/B is specified, paragraphs 3.5.1 and 3.5.2 shall apply.

3.6 Tearing strength (slit tear) Type I. At least 80 percent of the specimens tested shall have tearing strength values not less than 28 pounds and none below 20 pounds when tested as specified in paragraph 4.3.

3.7 Stitch tear strength Type II. At least 80% of the specimens tested shall have stitch tearing strength values not less than 30 pounds when tested as specified in paragraph 4.3.

3.8 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.9 Chemical requirements. The leather shall conform to the chemical requirements in Table I when tested as specified in paragraph 4.3.

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TABLE I. Chemical requirements

CHARACTERISTICS	MINIMUM	MAXIMUM
Hexane Soluble material, percent 1/	4.0	16.0
Chromic oxide (1/) shall be calculated on a hide substance basis (1/), percent	4.0	7.5
Non-determined material, percent (Type I)1/	-	20.0
pH value	3.0	4.0
Total ash, percent (Type I) 1/	-	10.0

1/ Calculated on a moisture-free basis.

4. VERIFICATION

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

Quality conformance inspection (see paragraph 4.2)

4.2 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with the referenced documents.

4.2.1 Component and material inspection. The above, components and materials shall be tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable procurement documents.

4.3 End item testing. The methods of testing specified in ASTM, and as listed in Table II, shall be followed. All test reports shall contain the individual values utilized in expressing the final result. The lot shall be rejected if any one of the following conditions exists:

More than three specimens fail requirements for water penetration or slit or stitch tearing strength.

Any specimen fails requirements for water absorption, or staining.

Any composite test result fails to meet the specified requirements.

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TABLE II. End item tests

CHARACTERISTICS	REQUIREMENT PARAGRAPH	TEST METHOD	SAMPLE UNIT	COMPOSITION
Material identification	3.1	Visual	-	-
Tannage	3.1	Visual	-	-
Staining	3.2	D6012 1/ 3/	X	-
Fungicide	3.4	4.4	-	X
Water absorption (treatment A)	3.5.1	D6015 or D6014 2/ 3/	X	
Water penetration (treatment B)	3.5.2	D2098 or D2099 2/ 3/	X	-

TABLE II. End item tests (continued)

CHARACTERISTICS	REQUIREMENT PARAGRAPH	TEST METHOD	SAMPLE UNIT	COMPOSITION
Slit tearing strength (Type I)	3.6	D2212 3/	X	-
Stitch tearing strength Type II	3.7	D4705 3/	X	-
Hexane –Soluble Material	3.9	D3495 3/	-	X
Chromic oxide	3.9	D2807 3/	-	X
Non- determined material (Type I)	3.9	4.5.1	-	X
pH value	3.9	D2810 3/	-	X
Total ash (Type I)	3.9	D2617 3/	-	X

1/ Except that the flesh side of the leather shall be in contact with the color transfer pad.

2/ The test method used shall be referenced in the test report.

3/ Refers to ASTM standards.

4.3.1 Sampling Plan. The sampling plan for the required tests shall be as follows:

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LOT SIZE (SQFT)	SAMPLE SIZE
Up to 25,000 sq. ft.	15

4.4 Fungicide.

4.4.1 TCMTB. Testing to determine the concentration of Busan 30 L in the leather shall be conducted by using Buckman Laboratories standard test method for 2- (Thiocyanomethylthio) Benzothiazole Analysis of Leather by HPLC. This method may be obtained by contacting Buckman Laboratories at 1256 North McLean Blvd., Memphis, TN 38108 (800) 282-5626.

4.4.2 Ortho-phenyl phenol. Testing to determine chemical concentration in leather shall be conducted in accordance with Test I, Type I of the MIL-L-8067 specification.

4.4.3 Diiodomethyl para Tolylsulfone. The chemical concentration in the leather shall be conducted using the High Pressure Liquid Chromatography (HPLC) test method as follows:

4.4.3.1 Grinding. Grind the leather according to paragraph 6.3.7 of ASTM Method D2813 Standard Practice for Sampling Leather for Physical and Chemical Tests.

4.4.3.2 Stock Standard Preparation. Weigh accurately approximately 100mg of Diiodomethyl-para-tolylsulfone Internal Reference Standard (IRS), into a 100ml volumetric flask. Add approximately 50ml of acetonitrile, stopper and sonicate to aid dissollution (about 1 minute). Dilute to volume with acetonitrile.

4.4.3.3 Working Standards Preparation. Transfer 3ml aliquots of the Stock Standard Preparation in to 25, 50, 100, and 250ml volumetric flasks. Dilute to volume with acetonitrile. These standards contain approximately 12, 30, 60 and 120ppm of Diiodomethyl-para-tolylsulfone.

NOTE:

(1) A lower standard may be required for samples containing very low levels of Diiodomethyl para tolylsulfone.

(2) Solutions must be stored in darkness to prevent degradation. Minimize exposure to light during preparation and use. The Standard Preparations may be retained one month.

(3) Internal Reference Sample (IRS) is available from Chemtan Company, Inc., P.O. Box C, Exeter, NH 03833 Tel.: 603-772-3741.

4.4.3.4 Sample preparation. Accurately weigh approximately 3 to 5g of sample into a 150ml beaker. Add 75-100ml of acetonitrile. Cover samples with parafilm; mix 16 to 24 hours on magnetic stir plate. The following day, allow samples to settle; remove necessary amount of acetonitrile and inject Note: Samples may have to be diluted further to fall within standard curve.

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4.4.3.5 Testing equipment:

HPLC System with variable UV detector and integrating computer. Suitable syringe and glassware.

A micro-Bondapak C18 column (Waters Associates P/N 2734) or equivalent.

0.45-micron polycarbonate filters (Nuclepore Corporation, Catalog No. 111107) or equivalent.

Calculator capable of doing linear regression.

Reagents: Acetonitrile (Chromatographic grade).

4.4.3.5.1 Typical mobile phase. Mix 550ml of distilled water and 450ml of acetonitrile. Filter and degas the solution. The concentration may be varied to meet system suitability requirements within range 40% to 50% acetonitrile.

4.4.3.5.2 Typical chromatographic conditions:

Injection Volume	20 Microliters
Flow	2.5 ml/minute
Detector	235 nm

4.4.3.6 System suitability. Inject the highest working standard until 2 percent agreement is reached between two successive injections. Chromatograph for approximately 15 minutes. Determine that, after the solvent front, one impurity peak elutes before the major peak. Elution for monoiodomethyl-para-tolylsulfone is about 6 minutes and for Diiodomethyl-para-tolylsulfone is about 8.5 minutes.

4.4.3.7 Procedure. Inject each standard and sample preparation in duplicate allowing each injection to run for at least 15 minutes. Measure the peak response for the Diiodomethyl-para-tolylsulfone.

Note: The run time may have to be increased for samples with late eluting peaks. Perform a linear regression of peak response (area) versus concentration for each of the components.

For each component:

$$\text{Concentration (ppm) in leather} = \text{Concentration (ppm) from linear regression.} \times \frac{75}{\text{Sample wt(g)}}$$

4.5 Methods of inspection.

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4.5.1 Non-determined material. The percent non-determined material shall be calculated as follows:

Non-determined material, percent = 100-(A+B+C)

Where A = Percent Hexane extractable material determined as specified in Table II.
B = Percent ash determined as specified in Table II.
C = Percent Hide Substance determined as specified in ASTM D2868

Note: Non- determined material is reported on a moisture-free basis, and therefore the values A, B, and C used in the calculation are also on a moisture free basis.

5 PACKAGING.

5.1 Packaging. Commercial packaging practices shall be used when shipping the finished leather hides from subcontractor to prime contractor.

6 NOTES:

Intended use. The cattlehide leather will be used in the upper leather parts of general purpose footwear.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Treatment required (see paragraph 1.2).
c. DODISS. The specific issue of individual documents referenced (see paragraphs 2.2.1 and 2.3).
d. Color, will be specified in end item footwear document.
e. Levels of preservation and packing (see paragraph 5.1).
f. When palletization is required (see paragraph 5.1).
g. When fungicides are not required (see paragraph 3.4).

6.3 Leather, cattlehide, gusset, chrome tanned, fatliquored Type II. The type II in this specification supersedes MIL-L-10867E, Leather, Cattlehide, Gusset, Chrome Tanned, Fatliquored (see paragraph 1.2).

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.5 Subject term (key word) listing
Footwear
Leather

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Custodians:

Army – GL
Navy – NU
Air Force – 11.6

Review activities:

Air Force – 99

Preparing activity:
DLA-CT

Project 8330-0198

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7, and send to preparing activity.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification

I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-PRF-3122J	**2. DOCUMENT DATE (YYYYMMDD)**

DOCUMENT TITLE
LEATHER, CATTLEHIDE, FOR FOOTWEAR UPPERS AND GUSSET, CHROME TANNED, FATILIQUORED

5. REASON FOR RECOMMENDATION

6. SUBMITTER		
a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON	**7.DATE SUBMITTED** (YYYYMMDD)

8. PREPARING ACTIVITY	
NAME Defense Supply Center Philadelphia ATTN: DSCP-CRFD (BLDG 6, AREA D)	b. TELEPHONE *Include Area Code)* (1) Commercial 215 737 7958 (2) AUTOVON 444-7958
c. ADDRESS *(Include Zip Code)* 700 Robbins Ave Philadelphia, PA 19111- 5096	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Standardization Program Office (DLSC-LM) 8725 John J. Kingman road, Suite 2533 Ft. Belvoir, VA 22060-2533 Telephone (703) 767-6888 AUTOVON 427-6888

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